

The size of the source image is O_{ij} with 127 pixel $\times$ 127 pixel. The center of the source image is defined as (0,0). To calculate the distorted image from the source image, we first calculate the discrete distortion kernel

$$K_{ijmn} = \exp\left(\frac{1}{2\sigma^2}(((1+f)m-i)^2 + ((1+f)n-j)^2)\right) \quad (1)$$

with $f = a_1(i^2 + j^2) + a_2(i^2 + j^2)^2$. a_1, a_2 are polynomial parameters which can be altered. σ is the parameter for a gaussian reference kernel. i, j, m, n are from -63 to 63 . Instead of calculating an integral, we calculate a discrete sum given by

$$I_{mn} = \sum_{ij} K_{ijmn} O_{ij} \quad (2)$$

It is convenient to run a simulation using MATLAB codes. The related function is mainly **FExampleP.m**. It receives 6 individual variables which are

- **imname**: string, name of the source image.
- **nname**: string, name of the output image.
- **p**: 1x2 double array, distortion parameter.
- **noise_ctrl**: string, receives only 'true' or 'false'. Determines whether the simulation includes poisson noise.
- **noise_parameter**: double, lambda of poisson noise. Useful only **noise_ctrl**='true'.
- **Noise_scale**: double, scale of noise. Useful only **noise_ctrl**='true'.

The recommended input image is updated from the 127 pixel $\times$ 127 pixel one to a 501 pixel $\times$ 501 pixel **pointsr.png**. Distortion parameter **p** is recommended to be $(-2, -2) \times 10^{-11}$.